$E = 200\text{V}$, with zero resistance. Motor resistance is $R = 10\Omega$; its no-load current is $I_0 = 2.0\text{A}$. For a steady mean cargo flow (mass conveyed per unit time) of $\eta = \frac{640}{9}\text{kg/s}$, find the mean motor current. Apart from cargo-belt friction and Joule heat, assume that all other energy losses are independent of the cargo load on the belt.

III. (20 points) A light rigid rod pivots about a smooth horizontal axis through O, freely in a vertical plane. Balls 1 and 2 on the rod are point masses $2m$ and m. Static friction has coefficient $\mu = \frac{5\sqrt{3}}{6}$. The rod is horizontal at rest; balls 1 and 2 are next to ends A and B, respectively. Release the system from rest. Find:

1. where ball 1 leaves the rod, as its angle θ_1 to the horizontal when ball 1 leaves;
2. where ball 2 leaves the rod, as its angle θ_2 to the horizontal when ball 2 leaves.

